

FANATIC CODERS OPERATIONS PLATFORM

Product Requirements Document (PRD)

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Prepared for: Fanatic Coders (fanaticcoders.com)

Classification: Internal — Confidential

1. Executive Summary

The Fanatic Coders Operations Platform (FCOP) is an internal operations backend with a client-facing portal, purpose-built for Fanatic Coders, a full-service digital agency based in Dubai offering Google Ads management, SEO, web development, mobile app development, and general marketing services.

The platform consolidates five currently fragmented workflows into a unified system: team and task management, client onboarding, project delivery tracking, client communication, and revenue operations. It integrates with Google Workspace (Chat and Calendar) to keep the team notified in real time, and provides clients with a branded portal where they can track progress, receive updates, share files, and view billing status.

V1 is scoped for fewer than 50 team members and 200 active clients. The platform is self-hosted, mobile-responsive (no native app in V1), and tech-stack agnostic at the PRD level.

Key Objectives

- Eliminate context-switching between 4+ separate tools (project management, CRM, invoicing, chat)
- Reduce average client onboarding time from 7–14 days to under 5 days
- Give clients real-time project visibility to reduce status inquiry emails by 60%+
- Provide admin with a single-pane revenue and operations dashboard
- Automate task notifications to Google Chat spaces to keep the team aligned

2. Problem Statement & Goals

2.1 Problem Statement

Fanatic Coders currently manages its operations across disconnected tools: project tasks in one platform, client communication via email and WhatsApp, invoicing in spreadsheets or separate billing software, and onboarding via ad-hoc Google Forms or documents. This fragmentation causes:

- Missed task handoffs and duplicated work due to lack of centralized task tracking
- Delayed client onboarding because information is gathered piecemeal across emails and calls
- Clients lack self-service visibility into project progress, generating repetitive status-check emails
- No unified revenue view — the admin cannot see revenue per client, per service, or per month without manual aggregation
- Internal notifications are informal (WhatsApp, verbal), leading to dropped tasks

2.2 Goals

#	Goal	Success Metric
G1	Centralize all operations in one platform	100% of active projects managed in FCOP within 90 days of launch
G2	Automate client onboarding per service type	Average onboarding completion < 5 business days
G3	Provide real-time client project visibility	> 70% of clients log into portal at least 1x/week
G4	Automate internal notifications via Google Chat	100% of task assignments trigger a Chat notification within 5 seconds
G5	Unified revenue and pipeline dashboard for admin	Admin can view MTD revenue, pipeline, and utilization without any manual data entry
G6	Reduce client status-inquiry emails	> 50% reduction in inbound status emails within 60 days of portal launch

3. User Personas

3.1 Admin (Agency Owner)

The agency owner who manages the business end-to-end. Needs visibility into revenue, team performance, client health, and pipeline. Has full permissions across all modules. Primary concern: business growth, profitability, and client retention.

3.2 Manager

Leads a team or service vertical (e.g., SEO Manager, Ads Manager). Assigns tasks, monitors team workload, communicates with clients, and oversees onboarding for their service line. Needs workload dashboards and the ability to manage projects and tasks within their purview. Cannot access revenue/billing data unless explicitly granted.

3.3 Team Member

Developers, marketers, SEO specialists, designers, and content writers. Executes tasks, logs time, uploads deliverables, and posts project updates visible to clients. Needs a clean task queue, clear priorities, and the ability to comment and attach files.

3.4 Client

External client who has engaged Fanatic Coders for one or more services. Interacts with the platform via the Client Portal. Needs to see project status, receive update notifications, reply to updates, download deliverables, and view billing status. Cannot see internal tasks, team workload, or other clients' data.

4. Feature Requirements

4.1 Module 1: Team & Role Management

Manages the internal team hierarchy, permissions, profiles, and workload visibility.

Feature	Priority	Description
Role hierarchy (Admin → Manager → Team Member)	P0	Three-tier role system with inheritance. Admin has full access; Manager scoped to assigned projects/teams; Team Member scoped to assigned tasks.
Permissions matrix	P0	Configurable CRUD permissions per role per module. Deny-by-default. Admin can customize Manager permissions.
Team member profiles	P0	Name, email, role, skills/tags, assigned projects, avatar. Editable by Admin and self (limited fields).
Utilization/availability tracking	P1	Shows each team member's current task load as a percentage. Based on assigned task hours vs. available hours per week (configurable per member).
Manager workload dashboard	P1	Aggregated view of team workload, task completion rates, overdue tasks, and upcoming deadlines for the manager's team.
Skill-based task suggestion	P2	When assigning a task, the system suggests team members based on matching skills and current availability.
Activity log per team member	P1	Chronological log of task completions, comments, and time entries. Visible to Admin and the member's Manager.

User Stories — Module 1

As a Admin, I want to define and customize role permissions per module, so that I can control exactly what Managers and Team Members can see and do, reducing security risk.

Acceptance Criteria:

- Admin can toggle CRUD permissions per role for each module
- Changes take effect immediately on next page load
- A denied permission hides the corresponding UI element and returns 403 on API call

As a Manager, I want to view my team's workload in a single dashboard, so that I can identify overloaded or underutilized team members and redistribute work.

Acceptance Criteria:

- Dashboard shows each team member's utilization percentage
- Tasks are color-coded by status (overdue = red, at risk = yellow, on track = green)
- Manager can click into a member to see their full task list

As a Team Member, I want to update my own profile with skills and availability, so that managers can assign me appropriate tasks.

Acceptance Criteria:

- Team Member can edit: skills tags, weekly available hours, timezone

- Team Member cannot edit: role, email, or assigned projects
- Changes are reflected in the skill-based suggestion engine within 1 minute

4.2 Module 2: Project Management

Core module for creating, tracking, and delivering projects and their constituent tasks.

Feature	Priority	Description
Project CRUD	P0	Create projects with: name, client, service type(s), status, start/end dates, budget, assigned team members. Edit and archive (soft-delete).
Task creation & assignment	P0	Tasks belong to a project. Fields: title, description (rich text), assignee(s), due date, priority (Low/Medium/High/Urgent), estimated hours, tags.
Customizable task statuses	P0	Default: To Do → In Progress → In Review → Done. Admin can add/rename/reorder statuses per project template.
Task dependencies	P1	Blocking and blocked-by relationships. A blocked task cannot move to In Progress until its blocker is Done.
Subtasks and checklists	P0	Tasks can have subtasks (each with assignee and status) and simple checklists (checkbox items).
Time tracking per task	P1	Team members can start/stop a timer or manually log hours. Time entries are linked to task, project, and team member for reporting.
File attachments per task	P0	Upload files (up to 25 MB per file, any common format). Files are stored per task and accessible to all assigned team members and the client (if marked as client-visible).
Task comments & mentions	P0	Threaded comments on tasks. @mention a team member to notify them. Supports rich text and file attachments in comments.
Kanban and list views	P0	Kanban board (columns = statuses) and flat list view with sort/filter. Both views per project.
Gantt/timeline view	P2	Visual timeline showing task durations, dependencies, and milestones. Read-only in V1; drag-to-reschedule in V2.
Recurring tasks	P2	Template-based recurring task creation (daily/weekly/monthly). Useful for SEO reporting, ad reviews, etc.
Project templates	P1	Save a project's task structure as a template. When creating a new project of that service type, auto-populate tasks from the template.

User Stories — Module 2

As a Manager, I want to create a project from a template when onboarding a new SEO client, so that the standard SEO deliverable tasks are pre-populated and I only need to assign team members and dates.

Acceptance Criteria:

- Manager selects a service-type template during project creation
- All template tasks are cloned with default statuses, priorities, and descriptions
- Manager can edit, add, or remove tasks before finalizing

As a Team Member, I want to log time against a task using a start/stop timer, so that my hours are accurately tracked for billing and utilization reporting.

Acceptance Criteria:

- Timer persists across page navigation (stored server-side)
- Only one timer can run per user at a time
- Time entries show in the task detail and in the team member's activity log

As a Admin, I want to view all projects in a portfolio view with filters for status, service type, and client, so that I can quickly assess the health of the agency's project pipeline.

Acceptance Criteria:

- Portfolio view shows project name, client, status, progress %, team lead, and budget consumed
- Filterable by: service type, status, client, date range, assigned manager
- Sortable by any column

4.3 Module 3: Google Workspace Integration

Connects FCOP to Google Chat and Google Calendar to keep the team informed in real time without requiring them to constantly check the platform.

Feature	Priority	Description
Google Chat task notification	P0	When a task is created or reassigned, automatically post a card-style message to a designated Google Chat space. Card includes: task title, assignee, deadline, priority badge, and a deep-link button back to the task in FCOP.
Google Chat space mapping	P0	Admin can map each project (or service team) to a specific Google Chat space. Notifications for that project go to the mapped space.
Google Calendar event creation	P1	Optionally create a Google Calendar event for tasks with deadlines. Event title = task title; event date = due date; description includes task link.
Status change notifications	P1	Post an update to Google Chat when a task moves to In Review or Done.
Daily digest to Chat	P2	Configurable daily summary posted to a Chat space: tasks due today, overdue tasks, tasks completed yesterday.

Integration Architecture

Authentication: The platform registers as a Google Chat App using a service account. This allows it to post messages to Chat spaces without per-user OAuth consent for each notification.

Required OAuth 2.0 Scopes:

- <https://www.googleapis.com/auth/chat.bot> — Send messages as the Chat app (primary scope for notifications)
- <https://www.googleapis.com/auth/chat.spaces.readonly> — List available spaces for the mapping UI
- <https://www.googleapis.com/auth/calendar.events> — Create/update Calendar events (user-delegated, requires one-time consent from the calendar owner)

API Endpoints Used:

- POST <https://chat.googleapis.com/v1/spaces/{space}/messages> — Create a message in a Chat space
- GET <https://chat.googleapis.com/v1/spaces> — List spaces the app has access to
- POST <https://www.googleapis.com/calendar/v3/calendars/{calendarId}/events> — Create a Calendar event

Constraints (from Google Chat API documentation):

- The Chat app must be added to a space before it can post messages to that space
- Messages created via app authentication display the app name as sender (not a user)
- Message size limit: 32,000 bytes per message

- App-specific OAuth scopes (chat.app.*) require one-time Workspace admin approval

User Stories — Module 3

As a Team Member, I want to receive a Google Chat notification when a task is assigned to me, so that I don't need to check the platform constantly to know I have new work.

Acceptance Criteria:

- Notification appears in the mapped Chat space within 5 seconds of task creation
- Card shows: task title, my name, due date, priority, and a clickable link to the task
- If the Chat space is not mapped, the system logs a warning and does not silently fail

As a Admin, I want to map each project to a Google Chat space, so that notifications for that project go to the right team channel.

Acceptance Criteria:

- Admin sees a dropdown of available Chat spaces (fetched via spaces.list)
- Each project can be mapped to exactly one space
- Mapping can be changed at any time; change takes effect immediately

As a Manager, I want to receive a daily digest of overdue tasks in my team's Chat space, so that I can take action on blockers first thing in the morning.

Acceptance Criteria:

- Digest is posted at a configurable time (default: 9:00 AM in the team's timezone)
- Lists: tasks overdue, tasks due today, tasks completed yesterday
- Each item links to the task in FCOP

4.4 Module 4: Client Onboarding System

A dynamic, multi-step onboarding workflow that adapts based on the service type(s) the client has engaged for. Industry research (ALM Corp, AgencyAnalytics, DesignRush, Leadsie) shows that agencies with structured onboarding achieve 70% higher retention and 31% faster time-to-revenue versus ad-hoc approaches.

Feature	Priority	Description
Dynamic service-type forms	P0	Onboarding form fields change based on selected service(s). A client engaging for both SEO and Google Ads sees combined relevant fields, not two separate forms.
Multi-step wizard UI	P0	Onboarding broken into logical steps: Company Info → Service-Specific Details → Access & Credentials → Goals & KPIs → Review & Submit.
Onboarding completion tracker	P0	Visual progress bar showing completion %. Admin/Manager can see which clients have incomplete onboarding and what's missing.
Save & resume	P0	Client can save partial progress and return later. Data persists across sessions.
Automated reminders	P1	If onboarding is incomplete after 48 hours, send an email reminder. Escalate to assigned Manager after 5 days.
Client account creation	P0	Upon onboarding initiation, the client receives an email invitation to create their portal login (email + password or magic link).
Onboarding data editable by Admin/Manager	P0	Internal team can view and edit all onboarding responses. Client can view but not edit after submission (can request changes).
File upload during onboarding	P0	Client can upload brand assets, design references, wireframes, existing reports, etc. during onboarding.
Secure credentials handling	P1	For third-party account access (Google Ads, GA4, GSC, social media), use permission-based access delegation (e.g., granting viewer/editor access) instead of sharing passwords. Include guidance text in the form.

4.4.1 Onboarding Field Specifications by Service Type

Common Fields (All Service Types)

- Company name, website URL, industry/vertical
- Primary contact: name, email, phone, timezone, preferred communication channel
- Secondary/billing contact (if different)
- Company size (employees), annual revenue range (optional)
- How did you hear about us? (attribution)
- Existing marketing tools/platforms in use
- Primary business goals (free text + structured checkboxes: brand awareness, lead gen, e-commerce sales, local traffic, etc.)
- Budget range for this engagement
- Timeline expectations (when do you need to see results?)
- Competitor URLs (up to 5)
- Brand guidelines upload (logo, color palette, fonts, tone of voice doc)

Google Ads — Service-Specific Fields

- Existing Google Ads account ID (or indicate new account needed)
- Current monthly ad spend
- Historical monthly ad spend (past 6 months if available)
- Target CPA / target ROAS
- Conversion actions currently tracked (list)
- Conversion tracking method (Google tag, GTM, offline import, etc.)
- Landing page URLs for current/planned campaigns
- Geographic targeting (countries, regions, cities, radius)
- Language targeting
- Negative keyword lists (upload or paste)
- Existing audiences/remarketing lists (Y/N, describe)
- Google Merchant Center ID (if Shopping campaigns)
- Product feed URL (if applicable)
- Previous agency/consultant? What worked/didn't work?
- MCC access delegation: provide admin email to grant access to Fanatic Coders MCC

SEO — Service-Specific Fields

- Primary domain (and any secondary/subdomain properties)
- CMS platform (WordPress, Shopify, custom, etc.)
- CMS admin access level available to the agency
- Google Search Console access (grant via email delegation)
- Google Analytics 4 property ID and access delegation
- Current organic traffic (monthly sessions, if known)
- Target keywords (if any; upload or paste list)
- Existing content inventory (blog posts, landing pages count)
- Content creation capability in-house (Y/N, capacity)
- Backlink profile: any past link-building or disavow activity?
- Known technical issues (site speed, indexing problems, manual actions)
- Local SEO needs (Google Business Profile access, NAP consistency)

- International SEO needs (hreflang, multi-language, multi-country)
- Previous SEO agency/consultant work (what was done, results)

Web Development — Service-Specific Fields

- Project type: new build, redesign, migration, feature addition
- Current website URL (if redesign/migration)
- Current hosting provider and plan
- Preferred tech stack (if any; e.g., React, WordPress, Shopify, Laravel)
- Domain registrar and DNS management access
- Design references / mood board URLs (up to 10)
- Sitemap or wireframes (upload)
- Number of pages/templates needed
- E-commerce requirements (product count, payment gateways, shipping)
- Third-party integrations needed (CRM, ERP, payment, analytics, chat)
- Content: who provides copy? (client, agency, or collaborative)
- Photography/media assets available (Y/N, upload)
- Accessibility requirements (WCAG level)
- Performance requirements (target load time, Core Web Vitals)
- SSL certificate status
- Post-launch support/maintenance expectations

Mobile App Development — Service-Specific Fields

- Target platforms: iOS, Android, or cross-platform
- Preferred framework (if any: React Native, Flutter, native Swift/Kotlin)
- Feature list / user stories (upload document or fill in)
- Design references / competitor app references
- Wireframes or prototypes (Figma/Sketch/Adobe XD links or uploads)
- Backend requirements: existing API, new backend needed, BaaS (Firebase, Supabase)
- Third-party API integrations (maps, payments, social login, push notifications)
- Apple Developer account (Team ID) and/or Google Play Console access
- Push notification requirements (provider preference)
- Offline functionality requirements
- User authentication method (email/password, social, SSO, biometrics)
- Expected user base size (DAU/MAU estimates)
- App Store Optimization (ASO) needs
- Analytics/crash reporting preferences (Firebase, Sentry, Mixpanel)

General Marketing — Service-Specific Fields

- Marketing channels currently active (social, email, content, PR, influencer, events)
- Social media account handles and admin access delegation
- Email marketing platform (Mailchimp, Klaviyo, HubSpot, etc.) and access
- Current email list size
- Target audience personas (upload or describe: demographics, psychographics, pain points)
- Campaign goals (awareness, engagement, conversions, retention)
- Seasonal/event-based marketing calendar (key dates)
- Existing brand assets (photo library, video library, templates)

- Content approval workflow (who approves, turnaround time)
- Influencer/partnership history
- PR contacts or agency (if separate)
- Regulatory/compliance constraints (industry-specific: healthcare, finance, etc.)

User Stories — Module 4

As a Client, I want to complete my onboarding by filling out a service-specific questionnaire online, so that the agency has everything they need to start working immediately.

Acceptance Criteria:

- Client receives an email invitation with a link to the onboarding wizard
- The form dynamically shows only the fields relevant to my engaged service(s)
- I can save progress and resume later from any device
- Upon submission, I see a confirmation and my assigned Manager is notified

As a Manager, I want to see which onboarded clients have incomplete submissions, so that I can follow up proactively and prevent onboarding delays.

Acceptance Criteria:

- Dashboard shows onboarding completion % per client
- Filterable by: service type, assigned manager, days since invitation
- One-click to send a reminder email to the client

As a Admin, I want to view onboarding analytics: average completion time, drop-off step, completion rate, so that I can identify and fix friction points in the onboarding flow.

Acceptance Criteria:

- Analytics show: average days to complete, % completed within 5 days, most-skipped fields
- Data is available per service type
- Exportable as CSV

4.5 Module 5: Client Portal

A branded, secure portal where clients can monitor their projects, receive updates, communicate with the team, and view billing status.

Feature	Priority	Description
Client authentication	P0	Email + password with email verification. Magic link login as alternative. Password reset flow. Optional: MFA (TOTP) for security-conscious clients.
Client dashboard	P0	Shows: active projects with current status, recent updates feed, upcoming milestones, and unread messages count.
Project status view	P0	Per-project view showing: current phase/status, milestone progress bar, recent task updates marked as client-visible.
Update feed with email notifications	P0	When a team member posts a project update (marked client-visible), the client sees it in the portal feed AND receives an email notification with a summary and link.
Client comment/reply on updates	P0	Client can reply to updates with text and file attachments. Reply is visible to the assigned team and triggers an in-app notification.
File sharing / deliverables	P0	Dedicated files section per project. Team uploads deliverables; client can download. Client can also upload files (revision requests, assets).
Invoice/billing visibility	P1	Read-only view of: invoices (amount, date, status: paid/due/overdue), payment history. Client cannot edit or pay through the portal in V1 (payment link included in invoice email).
Branded portal (white-label)	P1	Portal displays Fanatic Coders branding: logo, colors, favicon. No third-party branding visible to the client.
Multi-project support	P1	Clients with multiple engagements see all projects in one dashboard, with per-project drill-down.
Client satisfaction survey	P2	Periodic automated survey (CSAT/NPS) triggered after milestone completion. Results visible to Admin.

User Stories — Module 5

As a Client, I want to see the current status of all my projects on one dashboard, so that I don't need to email the agency to check on progress.

Acceptance Criteria:

- Dashboard shows each project with: name, status badge, progress %, last update date
- Clicking a project opens its detailed status view
- Data refreshes on page load (no stale cache beyond 60 seconds)

As a Client, I want to receive an email when the team posts a project update, so that I stay informed without having to log in daily.

Acceptance Criteria:

- Email is sent within 2 minutes of the update being posted
- Email includes: project name, update summary (first 200 characters), and a link to view full update
- Client can reply to the update directly from the portal (not from the email)

As a Client, I want to view and download my invoices, so that I can track what I owe and what I've paid.

Acceptance Criteria:

- Invoices section shows a list with: invoice number, date, amount, status
- Clicking an invoice shows the full invoice detail (line items, taxes, total)
- Download as PDF button available

4.6 Module 6: Admin Revenue & Business Dashboard

Gives the agency owner a comprehensive financial and operational overview in a single pane.

Feature	Priority	Description
Revenue tracking	P0	Dashboard widgets: total revenue (MTD, QTD, YTD), revenue per client, revenue per service type, revenue per month (12-month trend chart).
Invoice management	P0	Create invoices (line items, tax, discount, currency), send via email, mark as paid, overdue tracking. Overdue invoices auto-flagged at configurable thresholds (e.g., 7, 14, 30 days).
Pipeline view	P0	Kanban-style pipeline: Lead → Proposal Sent → Onboarding → Active → Completed → Churned. Each card shows client name, service type, estimated value.
Expense tracking	P1	Log expenses per project (third-party tools, ad spend pass-through, contractor costs). Links to project for margin calculation.
Profit margin per project	P1	Calculated as: $(\text{Revenue} - \text{Expenses} - (\text{Logged Hours} \times \text{Internal Hourly Rate})) / \text{Revenue}$. Requires time tracking and expense tracking to be active.
Client retention metrics	P1	Client retention rate (rolling 12 months), average client lifetime, churn rate, revenue per retained client.
Team utilization rate	P1	Aggregate and per-member: billable hours / available hours. Filterable by date range and service type.
Export and reporting	P1	Export dashboard data as CSV or PDF. Scheduled email reports (weekly/monthly) to Admin.

User Stories — Module 6

As a Admin, I want to see this month's revenue broken down by service type, so that I can identify which service lines are growing and which need attention.

Acceptance Criteria:

- Revenue widget shows a bar chart with MTD revenue per service type
- Hovering shows exact amount and % of total
- Clicking a service type drills down to client-level revenue

As a Admin, I want to create and send an invoice to a client, so that billing is handled within the platform without switching to a separate tool.

Acceptance Criteria:

- Invoice creation form includes: client (dropdown), line items, tax configuration, due date, notes
- Preview before sending
- Send triggers an email to the client's billing contact with PDF attachment and a portal link

As a Admin, I want to view the client pipeline from lead to active, so that I can forecast revenue and identify bottlenecks in the sales-to-delivery handoff.

Acceptance Criteria:

- Pipeline shows card count and total estimated value per stage
- Drag-and-drop to move clients between stages
- Moving to Onboarding stage triggers onboarding invitation email

4.7 Module 7: Notifications & Communication Layer

Manages all internal and client-facing notifications across channels: in-app, email, and Google Chat.

Feature	Priority	Description
In-app notification center	P0	Bell icon with unread count. Notifications for: task assigned, task status changed, comment/mention, update posted, deadline approaching (24h), onboarding completed. Clicking navigates to the relevant item.
Email notifications (internal)	P0	Immediate emails for: task assignment, @mention. Configurable digest (daily/weekly) for: completed tasks, project status changes. Team members can opt in/out per notification type.
Email notifications (client-facing)	P0	Triggered by: project update posted (client-visible), onboarding invitation, onboarding reminder, invoice sent, invoice overdue reminder.
Google Chat integration (covered in Module 3)	P0	Task creation/assignment and status change notifications to mapped Chat spaces.
Notification preferences	P1	Per-user settings: toggle on/off for each notification type per channel (in-app, email, Chat). Admin can set org-wide defaults.
Email delivery service	P1	Use a transactional email provider for reliability. Recommendation: Postmark (superior deliverability for transactional email) or SendGrid (broader feature set). Must support SPF/DKIM/DMARC for the fanaticcoders.com domain.
Webhook support for external integrations	P2	Expose webhooks for key events (task created, project status changed, invoice paid) so Admin can connect to Zapier or custom integrations in the future.

User Stories — Module 7

As a Team Member, I want to configure which notifications I receive and on which channels, so that I'm not overwhelmed by irrelevant alerts but never miss critical ones.

Acceptance Criteria:

- Settings page shows a matrix: notification type (rows) × channel (columns: in-app, email, Chat)
- Each cell is a toggle (on/off)
- Defaults are set by Admin but overridable by the user (except mandatory notifications like security alerts)

As a Client, I want to receive an email whenever there is an update on my project, so that I'm always aware of progress without needing to log in.

Acceptance Criteria:

- Email is sent within 2 minutes of a client-visible update being posted
- Email includes project name, update preview, and a button linking to the portal
- Client can unsubscribe from non-critical notifications (not invoices) via email preferences link

As a Admin, I want to set organization-wide default notification preferences, so that new team members and clients start with sensible defaults.

Acceptance Criteria:

- Admin settings page shows a default matrix identical to the user settings
- New accounts inherit admin defaults
- Admin can lock certain notifications as mandatory (cannot be turned off by users)

5. Non-Functional Requirements

5.1 Security

Authentication

- Internal users: email + password with bcrypt hashing (min 12 rounds). Enforce minimum password complexity. Support MFA (TOTP) for Admin and Manager roles (mandatory for Admin, optional for others).
- Clients: email + password or magic link (time-limited, single-use token sent via email). Optional MFA (TOTP).
- Session management: short-lived JWTs (15-minute access tokens) with server-side refresh tokens (rotated on each use). Secure, HttpOnly, SameSite=Strict cookies.

Authorization

- RBAC with deny-by-default. Every API endpoint validates role and scope.
- Client data isolation: every database query scoped by client_id. Implement Row-Level Security (RLS) at the database layer if using PostgreSQL, as defense-in-depth in addition to application-layer checks (per OWASP Multi-Tenant Security Cheat Sheet).
- Never trust client-supplied tenant/client IDs; derive from authenticated session.
- Use non-sequential, non-guessable UUIDs for all resource identifiers.

Data Isolation Between Clients

- Shared database with mandatory client_id column on all client-scoped tables (single-tenant data model within a shared database).
- Application middleware enforces tenant context on every request (per OWASP recommendation).
- Automated cross-tenant access tests in the CI/CD pipeline.

Audit Logging

- Log all create, update, delete operations with: user ID, timestamp, resource type, resource ID, action, old value (for updates), IP address.
- Logs stored separately (append-only) and retained for minimum 12 months.
- Admin can search and export audit logs.

Other Security Measures

- HTTPS everywhere (TLS 1.2+ enforced)
- CSRF protection on all state-changing endpoints
- Rate limiting on authentication endpoints (e.g., 5 failed attempts = 15-minute logout)
- Input validation and parameterized queries (prevent SQL injection)
- Content Security Policy headers
- File upload validation: check MIME type, enforce size limits, scan for malware (ClamAV or equivalent)
- Encrypted storage for sensitive data at rest (client credentials references, API keys)
- Regular dependency vulnerability scanning (Dependabot / Snyk)

5.2 Performance

- Page load time: < 2 seconds on standard broadband (for all dashboard and list views)
- API response time: p95 < 500ms for CRUD operations; p95 < 2s for dashboard aggregation queries
- Google Chat notification delivery: < 5 seconds from trigger event
- Email notification delivery: < 2 minutes from trigger event
- Support concurrent use by 50 internal users and 200 client users without degradation

5.3 Scalability

- V1 target: < 50 team members, < 200 active clients, < 1,000 active tasks
- Database schema and API design should support 10x growth without architectural changes
- File storage: use object storage (S3-compatible) with per-project prefixes. Plan for 500 GB in year one.

5.4 Accessibility

- WCAG 2.1 Level AA compliance for the Client Portal (since it's external-facing)
- Keyboard navigable; screen reader compatible; sufficient color contrast
- Internal admin UI: best-effort WCAG compliance (full compliance in V2)

5.5 Reliability & Availability

- Target uptime: 99.5% (allows ~3.6 hours downtime per month for maintenance)
- Automated database backups: daily full backup, hourly incremental. Retention: 30 days.
- Disaster recovery: documented restore procedure tested quarterly

6. Data Model Overview

This section outlines the key entities and their relationships. This is not a full schema but provides enough detail for an architect to begin database design.

Entity	Key Fields & Relationships
User	id (UUID PK), email, password_hash, first_name, last_name, role (enum: admin/manager/member), avatar_url, timezone, weekly_hours, skills (JSON), is_active, created_at, updated_at
Client	id (UUID PK), company_name, website, industry, primary_contact_name, primary_contact_email, primary_contact_phone, billing_contact_email, status (enum: lead/onboarding/active/completed/churned), pipeline_stage, source, created_at
ClientUser	id (UUID PK), client_id (FK → Client), email, password_hash, magic_link_token, mfa_secret, last_login, is_active
Project	id (UUID PK), client_id (FK → Client), name, service_type (enum[]), status, start_date, end_date, budget, chat_space_id, template_id (FK → ProjectTemplate, nullable), created_by (FK → User)
ProjectMember	project_id (FK → Project), user_id (FK → User), role (enum: lead/member) — join table
Task	id (UUID PK), project_id (FK → Project), title, description (rich text), status (FK → TaskStatus), priority (enum), assignee_ids (FK[] → User), due_date, estimated_hours, parent_task_id (FK → Task, nullable for subtasks), sort_order, created_by, created_at
TaskStatus	id, project_id (FK → Project), name, sort_order, is_default, is_done
TaskDependency	task_id (FK), depends_on_task_id (FK) — blocking relationship
TimeEntry	id, task_id (FK → Task), user_id (FK → User), started_at, ended_at, duration_minutes, note
Comment	id, task_id (FK → Task), user_id (FK → User), body (rich text), is_client_visible, parent_comment_id (nullable), created_at
FileAttachment	id, entity_type (task/comment/onboarding/project), entity_id, filename, storage_path, size_bytes, mime_type, uploaded_by, uploaded_at
OnboardingSubmission	id, client_id (FK → Client), service_types (enum[]), status (enum: invited/in_progress/submitted/reviewed), data (JSONB), completion_pct, invited_at, submitted_at, reviewed_by
Invoice	id, client_id (FK → Client), project_id (FK → Project, nullable), number, line_items (JSONB), subtotal, tax, discount, total, currency, status (enum: draft/sent/paid/overdue/void), due_date, paid_at, sent_at
Expense	id, project_id (FK → Project), category, description, amount, currency, date, receipt_file_id (FK → FileAttachment, nullable), logged_by
Notification	id, user_id (FK), type (enum), title, body, link, is_read, channel (enum: in_app/email/chat), created_at
AuditLog	id, user_id (FK), action (enum), resource_type, resource_id, old_value (JSONB), new_value (JSONB), ip_address, timestamp
ProjectTemplate	id, name, service_type, task_templates (JSONB)
NotificationPreference	user_id (FK), notification_type, channel, enabled (bool)

7. Competitive Feature Gap Analysis

Features found in competing agency management platforms (Teamwork, Monday.com, ClickUp, HubSpot, Agency Handy, ManyRequests, SuiteDash, Productive.io) that are NOT currently in the FCOP requirements but are worth considering:

Feature	Found In	Suggested Priority	Notes
Service catalog with public ordering page	Agency Handy, ManyRequests	P2 / V2	A public-facing page where prospective clients can browse services, select packages, and initiate onboarding. Useful for productized service offerings.
Client request queue (unlimited requests model)	ManyRequests	P2 / V2	Clients can submit design or content requests through a structured form, and the team works through them in priority order. Common in design-as-a-service agencies.
Built-in proposal & contract generation	Moxie, SuiteDash, Productive.io	P2 / V2	Create proposals, get e-signatures, and convert to projects within the platform. Eliminates need for DocuSign / PandaDoc.
Stripe/PayPal payment integration	Agency Handy, ManyRequests, SuiteDash	P1 / V1.1	Allow clients to pay invoices directly through the portal via Stripe or PayPal. Currently FCOP only shows invoices read-only.
White-label email (custom SMTP)	Agency Handy	P1 / V1	Send all platform emails from a custom domain (e.g., noreply@fanaticcoders.com) with full SPF/DKIM/DMARC. Already noted in notifications module but worth emphasizing.
Design proofing / visual feedback	ManyRequests, ClickUp	P2 / V2	Clients annotate directly on design deliverables (images, PDFs, web pages) rather than describing changes in text.
Resource scheduling / capacity planning	Productive.io, Teamwork	P2 / V2	Visual calendar showing team member availability and scheduled work across projects. More advanced than the utilization % in Module 1.
Client-facing reporting dashboards	AgencyAnalytics, HubSpot	P2 / V2	Auto-generated performance reports (SEO rankings, ad performance, traffic) pulled from integrated data sources and displayed in the client portal.
SLA tracking	Teamwork	P2 / V2	Define response-time and resolution-time SLAs per client/project and track compliance.
AI-powered task estimation	ClickUp, Monday.com	P2 / V2	Use historical time-tracking data to suggest estimated hours for new tasks of similar type.

8. Open Questions

The following decisions need your input before development begins:

ID	Question	Context / Impact
OQ-1	Payment gateway integration in V1?	Should clients be able to pay invoices directly via the portal (Stripe/PayPal), or is the read-only invoice view sufficient for V1? This significantly affects scope.
OQ-2	Tech stack selection	What is the preferred backend framework (Node.js/Express, Laravel, Django, Go)? What frontend framework (React, Next.js, Vue)? Database (PostgreSQL recommended for RLS support)?
OQ-3	Hosting environment	Self-hosted on a VPS (e.g., Hetzner, DigitalOcean), or on cloud infrastructure (AWS, GCP)? This affects the deployment and DevOps module.
OQ-4	Google Workspace edition	Does Fanatic Coders use Google Workspace Business or Enterprise? The Chat API's app-specific scopes (chat.app.*) require Workspace admin approval. Confirm admin access availability.
OQ-5	Internal hourly rates for margin calculation	To calculate project profit margins, the system needs an internal cost rate per team member (or per role). How should this be configured: per-member, per-role, or a flat agency-wide rate?
OQ-6	Currency and tax handling	Does the agency bill in a single currency (AED/USD) or multiple? Are there tax requirements (UAE VAT)? This affects invoice module complexity.
OQ-7	Client portal branding scope	Should the portal be fully white-labeled (custom domain like portal.fanaticcoders.com, custom email sender), or is a branded subdomain sufficient for V1?
OQ-8	Third-party file storage	Store files on the same server, or use S3-compatible object storage (e.g., Cloudflare R2, AWS S3, MinIO)? Object storage recommended for scalability and backup.
OQ-9	Multi-language support	Is the platform English-only, or should V1 support Arabic (RTL) given the Dubai client base?
OQ-10	Integration with existing tools	Are there any tools currently in use (e.g., Slack, WhatsApp Business, Zoho, QuickBooks) that need integration in V1, or is Google Workspace the only integration target?

9. Phased Rollout Recommendation

9.1 V1 (MVP) — Estimated 10–14 Weeks

Core platform that replaces the most critical fragmented workflows.

- Module 1: Team & Role Management (all P0 features)
- Module 2: Project Management (all P0 features + project templates P1)
- Module 3: Google Chat task notifications (P0 features)
- Module 4: Client Onboarding (all P0 features)
- Module 5: Client Portal (all P0 features)
- Module 6: Revenue Dashboard (P0: revenue tracking, invoice management, pipeline view)
- Module 7: Notifications (all P0 features)
- Non-functional: authentication, authorization, data isolation, audit logging, HTTPS, basic monitoring

9.2 V1.1 (Fast Follow) — 2–4 Weeks After V1

- Time tracking (Module 2, P1)
- Google Calendar integration (Module 3, P1)
- Invoice/billing visibility in client portal (Module 5, P1)
- Stripe/PayPal payment integration (competitive gap, P1)
- Notification preferences (Module 7, P1)
- Automated onboarding reminders (Module 4, P1)
- Utilization and workload dashboards (Module 1, P1)

9.3 V2 (Expansion) — Quarter After V1.1

- Task dependencies and Gantt view (Module 2, P1/P2)
- Daily digest to Chat (Module 3, P2)
- Expense tracking and profit margins (Module 6, P1)
- Recurring tasks (Module 2, P2)
- Client satisfaction surveys (Module 5, P2)
- Webhook support (Module 7, P2)
- Service catalog / public ordering page (competitive gap)
- Design proofing / visual feedback (competitive gap)
- Resource scheduling / capacity planning (competitive gap)
- Skill-based task suggestion (Module 1, P2)
- Client-facing performance reporting dashboards (competitive gap)

10. Research Sources & References

Client Onboarding Best Practices

- ALM Corp, "Client Onboarding Checklist for Digital Agencies: 17-Step Framework" (2025) — 86% of customers cite onboarding as a retention factor
- AgencyAnalytics, "Client Onboarding: How to Win Trust & Retain Clients" (2025) — structured onboarding as a scalable SOP
- Leadsie, "Agency Client Onboarding Best Practices" (2025) — shorter onboarding = 31% revenue increase; 70% retention improvement
- DesignRush, "Effective Agency Client Onboarding" (2025) — 1-2 week onboarding timeline benchmarks

Competitive Platforms Analyzed

- Agency Handy — CRM, onboarding, white-label portal, Kanban, invoicing
- ManyRequests — client request queue, branded portals, time tracking, Stripe
- SuiteDash — all-in-one CRM with white-label portals
- Productive.io — resource scheduling, profitability analysis, Slack/Calendar integrations
- Teamwork, ClickUp, Monday.com — general project management with agency features
- HubSpot — CRM pipeline management, client-facing reporting

Google Chat API

- Google Developers: Authenticate and Authorize Chat Apps (developers.google.com/workspace/chat/authenticate-authorize)
- Google Developers: spaces.messages.create API Reference (developers.google.com/workspace/chat/api/reference/rest/v1/spaces.messages/create)
- Google Workspace Admin: Set Up App Authorization for Chat (support.google.com/a/answer/15137461)

Security

- OWASP Multi-Tenant Security Cheat Sheet (cheatsheetseries.owasp.org) — tenant isolation, IDOR prevention, cache scoping
- OWASP Authorization Cheat Sheet — RBAC vs ABAC, deny-by-default, least privilege
- OWASP Cloud Tenant Isolation Project — cross-tenant vulnerability patterns